JUNWHA HONG

Research Officer (First Lieutenant), Republic of Korea Air Force

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RESEARCH INTERESTS

Computer Architecture for Emerging Workloads; Heterogeneous/Reconfigurable Computing.

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) Mar 2020 – Feb 2024
B.S. in Computer Science and Engineering, ***summa cum laude***

GPA: 4.13/4.3 (98.3/100), Ranked 4th out of 296 students

PUBLICATIONS

[1] **How Far Can PIMs Replace GPUs in Energy-Efficient Multi-Turn Agent Serving?**
Junwha Hong, Guseul Heo, Hyunsu Ye, Jongse Park, Olivia Hsu
IEEE International Symposium on Workload Characterization (IISWC), 2026; **short paper**.

[2] **Understanding and Exploiting Diagonal Attention Sparsity in Autoregressive Image Generation**
Daeun Kim, **Junwha Hong**, Changhun Oh, Yoonsung Kim, Yoonhyeong Lee, Jongse Park
IEEE International Symposium on Workload Characterization (IISWC), 2026.

[3] **Tortoise and Hare Guidance: Accelerating Diffusion Model Inference with Multirate Integration**
Yunghee Lee, Byeonghyun Pak, **Junwha Hong**, Hoseong Kim
Conference on Neural Information Processing Systems (NeurIPS), 2025.

[4] **CMASan: Custom Memory Allocator-aware Address Sanitizer**
Junwha Hong, Wonil Jang, Mijung Kim, Lei Yu, Yonghwi Kwon, Yuseok Jeon
IEEE Symposium on Security and Privacy (S&P), 2025 ***(Acceptance rate: 14.3%)***.

RESEARCH EXPERIENCE

Independent Research Collaboration May 2025 – Present
CMU SysCALL · KAIST CASys

Advisors: Prof. Olivia Hsu and Prof. Jongse Park.

- **GPU–PIM provisioning tradeoff (1st author, IISWC'26):** Characterized six agent workloads and identified fresh-prefill demand and live-session KV capacity as key limits to GPU replacement. Proposed retaining GPU-evicted session KVs on PIM to avoid recomputation.
- **Research infrastructure:** Extended LLMServingSim for long-context and memory-pressure studies, including a 5.3× faster analytical communication backend with approximately 2% error over Astra-Sim. Built a configurable simulator for rapid workload–mapping–hardware design-space exploration.
- **Sparsity in image generation (2nd author, IISWC'26):** Proposed the spatial-locality hypothesis behind diagonal attention and identified extreme sparsity as the key evaluation regime; contributed to evaluation and writing.

Agency for Defense Development (ADD) Mar 2024 – Present
Research Officer (First Lieutenant)

UNIST S2Lab Dec 2020 – Nov 2024
Undergraduate Researcher

Advisor: Prof. Yuseok Jeon.

- **CMASan (1st author, IEEE S&P'25):** Analyzed custom memory allocators across 100+ real-world C/C++ projects and generalized AddressSanitizer false negatives into recurring spatial and temporal patterns.
- Designed the compiler/runtime infrastructure for CMASan using Clang AST analysis, LLVM instrumentation, and allocator-aware runtime support, achieving 1.09× runtime overhead on average.
- Discovered 19 previously unknown memory bugs, including 7 CVEs, and contributed findings and fixes to projects including PHP, SQLite, MicroPython, libSQL, Google OSS-Fuzz, and AFL++.

University of Virginia Apr 2023 – May 2023
Visiting Student Researcher

Advisor: Prof. Yonghwi Kwon.

- Adapted CMASan to the OSS-Fuzz environment and evaluated it by fuzz-testing NGINX, Apache, gRPC, MySQL, and other applications.

Honors

Top Prize (Minister of Science and ICT Award) 2022
Millitech Challenge, Republic of Korea.

National Science & Technology Scholarship 2022

Excellence Prize 2021
UNIST-POSTECH-KAIST Data Science Competition.

Technical Skills

Languages: C/C++, Python, Rust, Scala

Compilers & Analysis: LLVM/Clang, TVM, ONNX/ONNX Runtime, CodeQL

ML Systems & Profiling: vLLM, SGLang, NVIDIA Nsight

Hardware & RTL: Chisel, Vivado HLS; NVIDIA A100/H100/H200/B300, FPGAs

References

To be added after confirmation.